Pseudonym Resolution and Formal Verification of an Unlinkable Two-Message IoT Authentication Protocol

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***Abstract*—In many lightweight IoT authentication protocols, a device sends an unlinkable, epoch-rotated pseudonym in clear and encrypts everything else under a key that depends on the device. A server therefore has to find out which device sent a message before it can decrypt or verify it. We study this resolution step for the two-message protocol of Al-Rasheed et al. (IEEE TCE, 2026). From the specification alone, we show that every correct server needs the step, and that the protocol neither specifies it nor costs it. We implement the exchange in Python, check it against independent reference vectors, and add seven resolution strategies: four that keep the wire format and three from RFID privacy research (randomized hash locks, key trees and rainbow time-memory trade-offs). Probe counts are measured on the implementation and checked against closed forms. They are then priced with the paper's own per-operation constants in an OMNeT++ model of its stated network. At one device every strategy reproduces the paper's 23.68 ms per session. At 1,000 devices with 5% adversaries, a precomputed table keeps the session at 24.4 ms. The on-demand scan that the specification suggests reaches 44 s, because each forged pseudonym forces a full scan that the block-list never sees. Counter epochs also lock a device out for good after eight dropped messages. A ProVerif analysis shows that session-key secrecy and agreement hold for every correct resolver. Unlinkability holds with counter epochs and fails with clock epochs, and injective authentication of the device needs the replay state that the specification leaves open. Code, models and data are open.**

***Index Terms*—IoT authentication, pseudonym resolution, unlinkability, private identification, scalability, OMNeT++, formal verification, ProVerif**

I. Introduction

Lightweight mutual-authentication protocols for IoT devices often hide the device's identity behind a pseudonym that changes over time, so that an eavesdropper cannot link two sessions of the same device. Al-Rasheed et al. [1] propose such a protocol. Devices and servers share pairwise keys from an offline phase, and they authenticate in two messages using symmetric primitives only. The device sends $PID = H_2(\lambda_V \| t)$ in clear. The epoch t and the nonce N_i travel inside a ciphertext under the pairwise key $k^\star$.

This design has a cost on the server side. The server cannot decrypt the first message until it knows which device sent it, and the pseudonym is designed to hide exactly that. RFID privacy research has studied this tension for twenty years [2]–[7]. It is known as *private identification*: without precomputation, a reader that must identify one of N tags from an unlinkable response does $O(N)$ work. In the protocol of [1], the step is not specified, and the cost model (its Table V and Eq. 22) has no term for it. Its evaluation assigns the server a constant $2T_H + 2T_{SE/D}$ per session, whatever the number of devices.

This paper makes the following contributions:

1) *The resolution gap.* From the specification alone, we show that every correct server must resolve the pseudonym before any cryptographic check, and we state which choices the specification leaves open (Section III).
2) *A design space.* We map seven strategies along the two open choices, the epoch semantics and the resolution method: four that keep the wire format, and three adapted from RFID work that change the first message. Each has a closed-form or measured cost (Section IV).
3) *A ranked comparison under the paper's own conditions.* Probe counts measured on an open implementation are priced with the paper's Table III constants. They are evaluated in an OMNeT++ model of its Table II network,

and the one-device column is taken from the paper itself (Section VI, Table II).

4) *Failure modes.* We show the denial-of-service and desynchronisation behaviour of each strategy, including a permanent lockout after eight dropped messages, which the paper's threat model allows.
5) *A formal analysis.* ProVerif models establish which security properties are independent of the resolver, and which depend on the epoch choice (Section VII).

We do not re-evaluate the other results of [1]. We take its protocol as the specification and build on the parts that we reproduced exactly.

II. Background

A. The protocol

In the offline phase, a trusted authority derives each entity's long-term secret $\lambda_X = H_1(ID_X \| s)$ and gives each device–server pair the pairwise key $k^\star = \mathrm{KDF}(\lambda_V \| \lambda_R)$ (Eqs. 8–9 of [1]). Online, in epoch t, the device V samples N_i and sends

$$M_1 = \langle PID,\ \mathrm{AEAD}_{k^\star}(PID \| N_i \| t) \rangle, \quad PID = H_2(\lambda_V \| t).$$

The server resolves PID to a device and an epoch, decrypts and verifies M_1, samples N_j, and returns $M_2 = \langle ID_R, \mathrm{AEAD}_{k^\star}(N_j \| t;\ PID \| N_i) \rangle$. Both sides derive $K = \mathrm{KDF}(k^\star \| N_i \| N_j \| t)$ (Eqs. 11–20). We instantiate AEAD as AES-256-GCM [13], KDF as HKDF-SHA256 [11] and H as SHA-256 [12], as the instantiation notes of [1] suggest.

Our Python implementation of this exchange, and a C port of its device side, reproduce reference vectors byte for byte (15 checks each). The vectors were computed by a generator that shares no code with the implementations. They cover the pairwise key, the pseudonym, the wire bytes of M_1 and M_2, and the session key, together with the rejection of tampered and replayed messages. The vectors fix the message layout that the next section relies on: the pseudonym is the only field in clear, and N_i and t are inside the AEAD ciphertext under $k^\star$.

B. Private identification in RFID

The same problem appears when an RFID reader must identify a tag that answers with an unlinkable value. Four lines of work define the strategies we adapt:

- *Randomized hash lock* [2]: the tag answers $(r, H(k \| r))$ for a fresh r, and the reader tries every key, $O(N)$.
- *Hash chains* (OSK) [3]: the tag's state evolves by a one-way function, which gives forward privacy. The reader searches N chains up to depth m, $O(Nm)$. Our counter epochs are the same search with the chain replaced by $H_2(\lambda \| t)$.
- *Key trees* [4]: keys form a tree of branching factor δ, and the tag answers with one keyed hash per level, $O(\delta \log_\delta N)$. Tags share their internal keys, so compromising one tag weakens the privacy of others [6].
- *Time-memory trade-off* [5], [6]: Hellman-style [8] rainbow tables [9] over all (tag, state) pairs bring OSK to $O((Nm)^{2/3})$ time and memory.

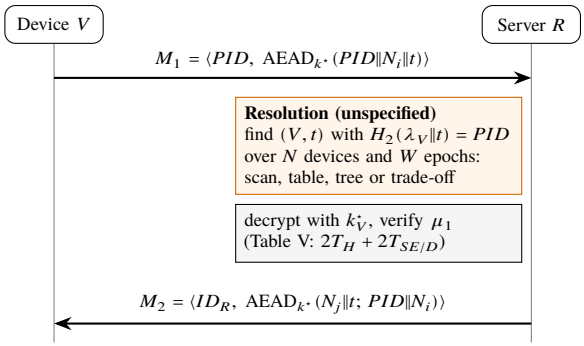


Fig. 1. Where resolution sits. The server needs $k^\star_V$ to open M_1, but V is hidden behind PID.

Time-based pseudonyms, as in YA-TRAP [7], let the reader precompute one table per time period. Avoine et al. [6] compared these approaches by time and memory for one tag. Our Table II makes the same comparison for this protocol, at network scale and under the paper's own constants.

III. The Resolution Gap

Each step of the argument uses only the specification, or a fact that our reference vectors confirm.

1) M_1 carries PID in clear. The epoch t and nonce N_i are only inside the ciphertext under $k^\star$ (Eqs. 13–15; confirmed byte for byte).
2) $k^\star = \mathrm{KDF}(\lambda_V \| \lambda_R)$ is specific to the device (Eqs. 9, 12).
3) So the server cannot decrypt M_1 or verify μ_1 until it knows which device sent it.
4) The paper requires that PID cannot be linked to V without λ_V (its Section III-J). So the device's identity cannot be read from M_1 either.
5) So every correct server must map PID to (V, t) by computing pseudonyms from the secrets it holds. We call this step *pseudonym resolution*.

The specification says only that the pseudonym is "rotated with time (or epoch) t". It leaves two choices open, and together they decide the cost:

- *Epoch semantics*: a per-device counter, or a clock.
- *Resolution method*: an on-demand search, or a table prepared in advance.

Table V of [1] charges the server $2T_H + 2T_{SE/D}$ per session, and Eq. 22 lists one KDF, one encryption and one MAC. Neither contains a term that grows with the number of devices. We count resolution cost in *probes*, where one probe is one evaluation of H_2 (or the equivalent keyed hash), and price it at T_H from the paper's Table III.

IV. Design Space

We implemented seven strategies (Table I). The first five keep the wire format of M_1. The last two need a fresh r in clear, so they change M_1. They are modelled at the resolution level, with the same keyed hash.

- **O1**, counter with a linear scan, is the behaviour of our original implementation. For each device it probes epochs

$\ell - L$ to $\ell + F$, where ℓ is the last accepted epoch, with $L = 16$ and $F = 8$ (window $W = 25$).

- **O2** keeps a table of every device's next F pseudonyms and slides it after each acceptance.
- **O3** and **O4** use clock epochs with a drift window $W_t = 3$. O3 scans, and O4 extends a table once per epoch.
- **TM** is a rainbow table over $S = Nm$ points, with lifetime $m = 128$ epochs per device, chain length $\lceil S^{1/3} \rfloor$, $2S^{2/3}$ chains per table, and six tables.
- **CR** and **MW** use $\delta = 32$, so the tree has two levels up to $N = 1024$.

Validation. For each strategy and each $N \in \{1, 2, 5, \dots, 10^4\}$ (15 values), a script builds the server in steady state. It resolves 200 pseudonyms of randomly drawn devices (50 for the scanning strategies above $N = 2{,}000$) and 5 forged ones, and counts every H_2 call. For O1–O4 and CR, the forged and per-epoch counts and the state equal the closed forms exactly. O2 and O4 need exactly 1 and 0 probes per authentication. The honest means of O1, O3 and CR lie within four standard errors of the formulas. MW never exceeds δd. In our parameter runs ($N = 40$ to 500), the TM tables covered 99.6–99.8% of points, and in the sweep 9 of 2,100 honest pseudonyms were missed; the device then retries with its next epoch. Twelve unit tests run each wire-compatible strategy through the full handshake. They check its probe counts, replay rejection and lockout behaviour, and they show that two sessions in one clock epoch share a pseudonym. All 125 tests pass.

V. Evaluation Method

We follow the paper's evaluation conditions and change only the resolution term.

- *Simulator and environment*: OMNeT++ [10] 6.4 (Cmdenv), with Table II of [1]: a 1500×1500 m area, 10 Mbps, 500 m range, 75.6 mW, a 15 μs quantum, servers at 5% of the devices.
- *Protocol cost*: Tables III–V of [1]. That is $2T_H + 2T_{SE/D} = 11.84$ ms per side, 1,024 bits over four messages, and $T_H = 0.32$ ms per probe.
- *Resolution cost*: for each server we look up the probes for its number of enrolled devices n in the measured cost model. It is exact for O1–O4 and CR, and interpolated between measured points for TM and MW. CR and MW also add their extra M_1 bits.
- *Assumptions* (the paper does not fix these): an adversary share of 5%, each sending forged M_1 at 1 Hz; a 10 s authentication window; each server enrols every device in its range; first-come-first-served server processing; for O4, one epoch extension per server inside the window.
- *Runs*: five seeds per point and $N \in \{1, 10, 100, 200, \dots, 1000\}$, with and without adversaries. That is 960 runs in total, reported as means.

The model is ours, not the authors'. With the resolution term removed, it gives 23.86 ms at $N = 1$ and 24.07 ms at $N = 1000$, against the paper's arithmetic value of 23.68 ms. The difference is airtime and quantisation. The *base* column of Table II ($N = 1$) is therefore the paper's own figure: 23.68 ms from Tables III and V, and 320 bits of server storage from Table VI. Storage at larger N is Table VI's 320 bits per enrolled device plus the resolution state, averaged over the servers of the simulated network.

VI. Results

A. Ranking

Table II gives the ranking. The two table strategies, O2 and O4, stay within 1 ms of the paper's figure up to 1,000 devices. O2 needs one probe to slide the window, and O4 only the per-epoch extension. Their cost is memory: at 1,000 devices, about 93 KB and 46 KB per server, against 10 KB for Table VI alone. The key tree is third (31.4 ms). It changes M_1, and a compromised device leaks the internal keys it shares with others. The randomized hash lock (CR) needs no extra state, but its latency grows linearly, to 95 ms. The three scanning strategies with epochs saturate the server: clock scan (O3) at 1.1 s, the time-memory trade-off at 12.9 s, and the counter scan (O1) at 44 s. For O1 and TM, the spread across seeds at $N = 1000$ is 10.6 s and 4.5 s, because the latency depends on how many devices and adversaries share the busiest server.

B. Honest load against forged load

Fig. 2 separates the two causes. Without adversaries (a), O1 already reaches 19.2 s at 1,000 devices. Each honest authentication costs about $12.5\,n$ probes, which is 1.0 s of hashing at the $n \approx 258$ devices that a server enrols at $N = 1000$ ($T_H = 0.32$ ms). Arrivals outpace this, so the server queue grows. The forged load (b) roughly doubles this, because every forged pseudonym costs a full miss scan of Wn probes. Such a pseudonym never resolves to a device, so the block-list, which counts failures per device, never sees it. Fig. 3 shows the same effect for one server up to 10^4 devices. A forgery costs O1 250,000 probes and TM about 46,000, but costs O2 and O4 nothing, because a table miss needs no hash.

C. Time against memory

Fig. 4 shows storage at one server, and each strategy's position in the time–memory plane at $N = 1000$. No strategy that keeps the wire format is both fast and small. O2 buys constant time with F table entries per device (328 KB in total), and O4 with W_t entries per device (144 KB). TM sits in between, at 131 KB and 1,478 probes. It is dominated here because its table must cover the device's whole lifetime ($m = 128$ epochs), not just the current window. MW is the only point that is both small and fast, and it pays with a different M_1 and a weaker privacy model.

D. Desynchronisation

With counter epochs, the device advances t every time it builds M_1. The server searches at most F epochs ahead of the last one it accepted. In our implementation, an adversary who drops $F = 8$ consecutive M_1 messages locks the device out for good: every later handshake fails with an unknown pseudonym, with or without an earlier successful session. The

TABLE I

DESIGN SPACE. N: DEVICES ENROLLED AT THE SERVER; $L = 16$, $F = 8$, $W = L + F + 1$; $W_t = 3$; ONE TABLE ENTRY IS 40 B. "PROBES" ARE EXPECTED H_2 EVALUATIONS PER MESSAGE, IN STEADY STATE.

	Epoch / M_1	Probes, honest	Probes, forged	Precompute	Resolution state	Dropped M_1	Linkability
O1	counter / unchanged	$\frac{(N-1)W}{2} + L + 2$	NW	none	$8N$ B	lockout after F	none
O2	counter / unchanged	1 (slide)	0	NF once	$(40F + 8)N$ B	lockout after F	none
O3	clock / unchanged	$\frac{(N-1)W_t}{2} + 1$	NW_t	none	nonce cache	none	same epoch
O4	clock / unchanged	0	0	N per epoch	$40W_tN$ B + cache	none	same epoch
TM	counter / unchanged	≈ 0.5–$0.7\,S^{2/3}$ (meas.)	$\approx 4\,S^{2/3}$ (meas.)	$\approx 12S$ once	$\approx 50\,S^{2/3}$ B	none until $t = m$	none
CR	none / $+r$ (128 b)	$\frac{N+1}{2}$	N	none	none	none	none
MW	none / $+r$, $+(d-1)$ tags	$\leq \delta d$	$\leq \delta$	none	32×internal nodes	none	after compromise

TABLE II

STRATEGIES RANKED FROM BEST TO WORST BY MEAN SESSION LATENCY AT $N = 1000$ (5% ADVERSARIES). TIME: MEAN PER-SESSION LATENCY IN THE OMNET++ MODEL OF THE TABLE II NETWORK, TABLE III CONSTANTS, MEAN OF FIVE SEEDS. SPACE: MEAN STORAGE PER SERVER. N IS THE NUMBER OF DEVICES IN THE NETWORK; AT $N \geq 100$ EACH SERVER ENROLS ABOUT A QUARTER OF THEM. THE BASE COLUMN ($N = 1$) IS THE VALUE PUBLISHED IN [1]: 23.68 MS (TABLES III, V) AND 320 BITS (TABLE VI).

			Complexity per auth.		Mean session latency (ms)					Storage per server (KB)				
#		Strategy	Time	Space	base	10	100	500	1000	base	10	100	500	1000
1	O2	Counter + precomputed table	$O(1)$	$O(NF)$	23.68	24.17	24.25	24.40	24.40	0.04	0.93	8.78	46.1	92.9
2	O4	Clock + per-epoch table [7]	$O(1)$, $+O(N)$/epoch	$O(NW_t)$	23.68	23.85	23.93	24.22	24.59	0.04	0.47	4.39	23.1	46.4
3	MW	Key tree [4]	$O(\delta \log_\delta N)$	$O(N)$	23.68	24.44	27.96	30.86	31.38	0.04	0.10	0.97	5.16	10.4
4	CR	Randomized hash lock [2]	$O(N)$	$O(N)$	23.68	24.44	28.86	51.71	94.89	0.04	0.10	0.95	5.02	10.1
5	O3	Clock + linear scan	$O(NW_t)$	$O(N)$	23.68	24.94	38.99	176.1	1,108	0.04	0.16	1.53	8.03	16.1
6	TM	Rainbow time-memory trade-off [5], [6]	$O((Nm)^{2/3})$	$O((Nm)^{2/3})$	23.68	38.33	516.5	5,228	12,850	0.04	2.26	10.8	36.5	59.9
7	O1	Counter + linear scan [3]	$O(NW)$	$O(N)$	23.68	36.02	507.9	16,947	44,010	0.04	0.12	1.15	6.02	12.1

lockout holds for O1 and O2, and a unit test checks it for both. A device whose counter lives in RAM is also rejected as a replay after a restart; we observed this for O1. Message dropping is within the threat model of [1]. Clock epochs (O3, O4), CR and MW have no per-device counter, so they cannot desynchronise this way. TM tolerates any number of drops until t reaches the lifetime m.

VII. FORMAL VERIFICATION

We modelled the exchange in ProVerif 2.05 [14], in the symbolic model with an active attacker (Table III). Our hash, KDF and AEAD are free functions, and AEAD has associated data. The main model has unbounded servers, unbounded honest and corrupt devices per server, and unbounded sessions. A corrupt device's λ and $k^\star$ are given to the attacker. The server may resolve a pseudonym to *any* enrolled device, and must then decrypt and check $PID = H_2(\lambda_V \| t)$. So every result that holds, holds for every resolver that returns a correct device. Epochs are fresh values, and the server keeps no replay state. Unlinkability is checked as observational equivalence [15], in the style of [16]: one device running twice, against two devices running once each. On both sides the server knows both devices and resolves with one destructor.

These results separate the properties that the resolver cannot affect from those it decides:

- *Independent of the resolver*: secrecy of the session key (Q1, Q2), agreement on (V, R, N_i, t) (Q3), and injective key agreement for the device (Q5). Q5 holds because N_i is fresh and bound as associated data of M_2.

TABLE III

PROVERIF RESULTS (ALL MODELS FINISH IN UNDER 1 S).

Property	Result
Q1/Q2 Session key secret (device / server side)	true
Q3 Server accepts M_1 of honest $V \Rightarrow V$ sent it	true
Q4 Same, injective (no replay state)	false (replay)
Q5 Device accepts $M_2 \Rightarrow$ server derived the same key, injective	true
FS Key secret after later leak of $k^\star$	false
U1 Unlinkability, counter epochs	true
U2 Unlinkability, clock epochs, same epoch	fails (equal PIDs)

- *Decided by the epoch and replay choices*: injective authentication of the device (Q4) fails without replay state. ProVerif finds a trace in which one M_1 is accepted twice. Injectivity therefore rests on the epoch window (O1, O2, TM) or the nonce cache (O3, O4), which the specification leaves open. With clock epochs, the attacker links two sessions in one epoch by testing their pseudonyms for equality (U2). With counter epochs, the equivalence holds (U1).
- *Forward secrecy*: it does not hold, and the base paper does not claim it. A later leak of $k^\star$ reveals N_i, N_j and t from the ciphertexts, and with them the session key.

Q4 cannot be settled in ProVerif with the replay window included, because the window is mutable state. We check the window by test instead, and leave a stateful proof (for example

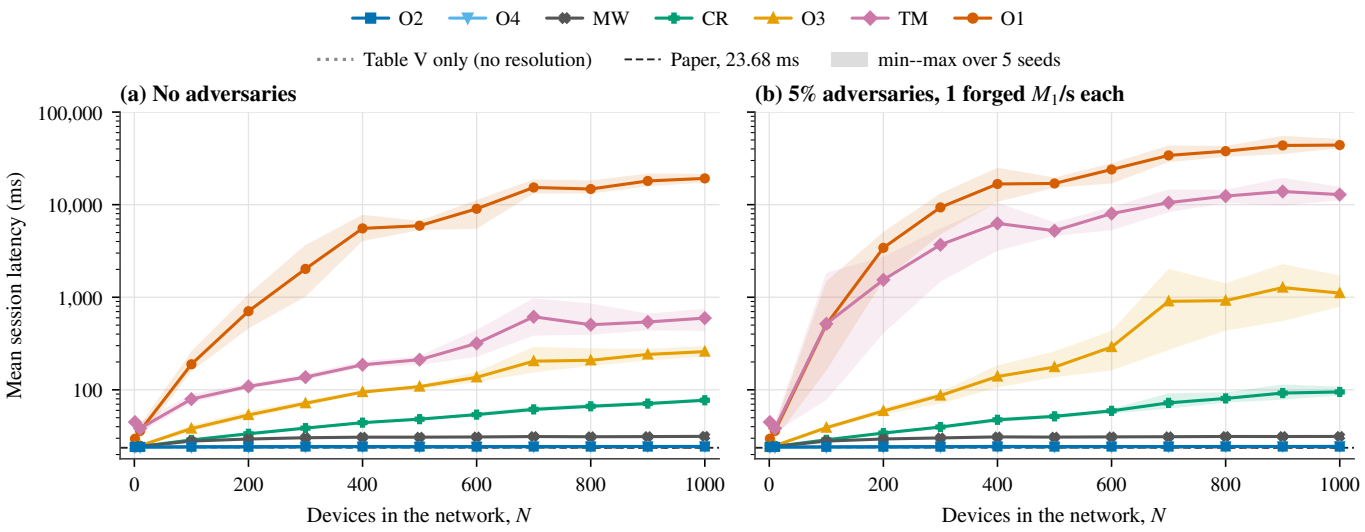


Fig. 2. Mean session latency against network size in the OMNeT++ model (log scale; band: minimum to maximum over five seeds). Dotted: the paper's cost model with no resolution. Dashed: the paper's 23.68 ms.

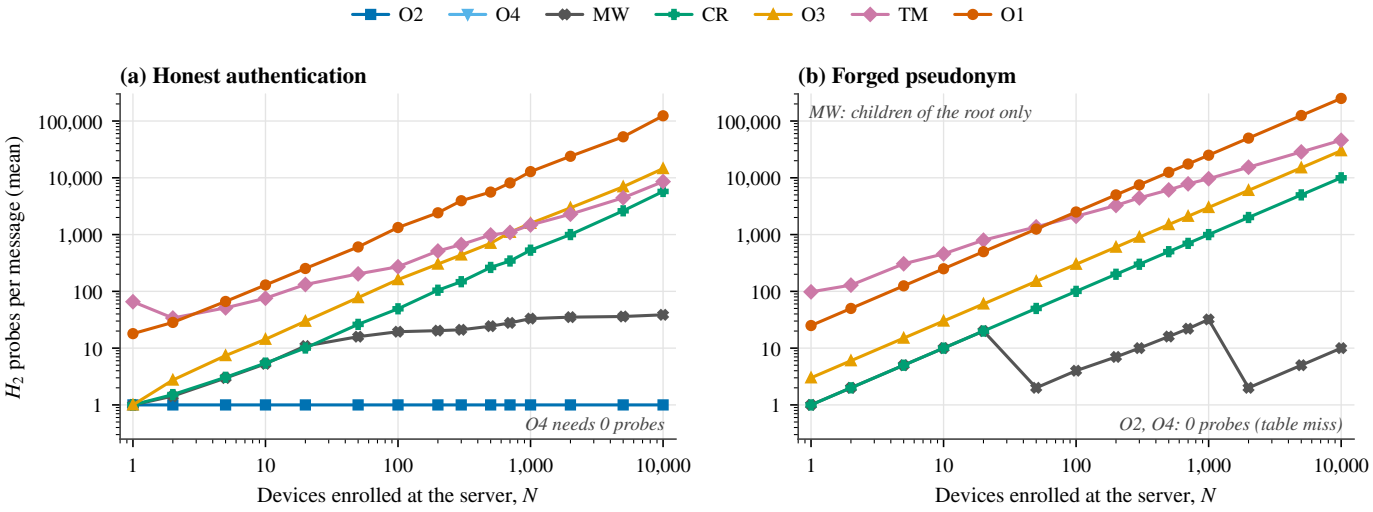


Fig. 3. Probes per message at one server, measured on the implementation (log-log). (a) Honest authentication. (b) A forged, random pseudonym.

in Tamarin) to future work.

VIII. Discussion

What a deployment should choose. A server that uses counter epochs should precompute (O2). One probe per authentication and no hashing on forged input remove both the scaling and the denial-of-service problem, at about $40F$ bytes per device. The lockout after F dropped messages remains, so the device needs a resynchronisation path, or the server needs a larger F, which costs table space linearly. If devices have clocks, O4 has the lowest latency and half the memory of O2. It makes sessions within one epoch linkable and needs a nonce cache for replay detection. The epoch length then trades privacy against the per-epoch rebuild of N probes. On-demand scanning (O1, O3) is acceptable only for small N and a trusted network. With 5% adversaries, O1 is already 21 times the paper's per-session cost at 100 devices.

What the specification should state. The protocol needs three decisions that it does not state: the epoch semantics, the resolution method, and the replay state that makes device authentication injective. Its cost model needs a resolution term. That term is $(N-1)W/2 + L + 2$ probes for an on-demand counter scan, and at least one probe or one table lookup for a precomputed table. Its storage model needs the table.

IX. Limitations

- The simulation is our model of the paper's stated environment, not the authors' simulation, which is not available. The adversary rate, the authentication window and the

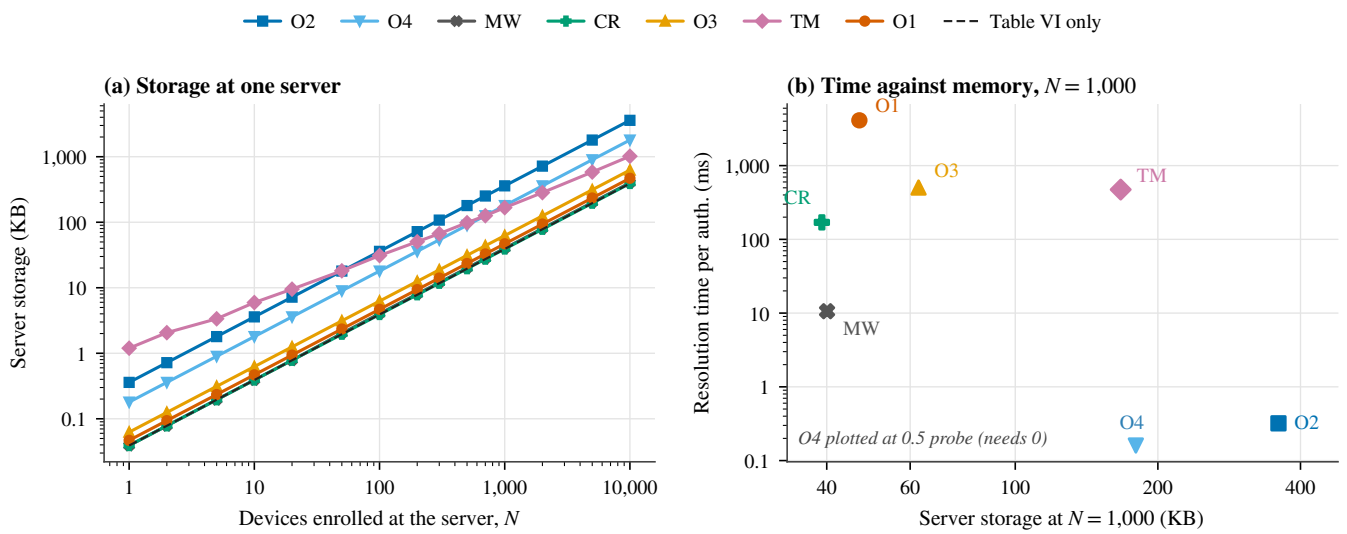


Fig. 4. (a) Server storage (Table VI's 320 bits per device plus the resolution state) for one server. (b) Resolution time per authentication (T_H = 0.32 ms per probe) against storage, at N = 1000.

enrolment rule are our assumptions, and the saturated latencies of O1, O3 and TM depend on them.

- We price probes with the paper's Table III constant (T_H = 0.32 ms) and report no hardware timings.
- CR and MW change M_1, and we model them at the resolution level only.
- The TM parameters (six tables, m = 128) are one point of its trade-off curve.
- Our epoch, window and table choices are ours, not the paper's.
- Conformance means agreement with our reading of Section III, which only the authors can confirm.
- The ProVerif results are symbolic and assume perfect primitives.

X. Conclusion

A server cannot authenticate a device that it has not yet identified. In the protocol of [1], that identification is a required step with no specification and no cost in the model. Under the paper's own constants and network, the choice of how to fill the gap decides whether the server stays at the paper's 23.68 ms per session or saturates at tens of seconds. It also decides whether a few dropped messages lock a device out, and whether forged pseudonyms are free to send. Precomputed tables, the oldest answer from RFID private identification, keep the protocol's cost claim intact at a memory cost of tens of kilobytes per server. The formal analysis shows that the protocol's secrecy and agreement properties do not depend on this choice, while unlinkability and injective authentication do. The implementation, the counting scripts, the OMNeT++ model, the ProVerif models and all results are open at https://github.com/jaykerkar0405/hybrid-ecc-iot-auth.

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